Aalto University
School of Arts, Design
and Architecture

Check point no.5
Topic: Distributed Mixture-of-Experts (MoE) systems

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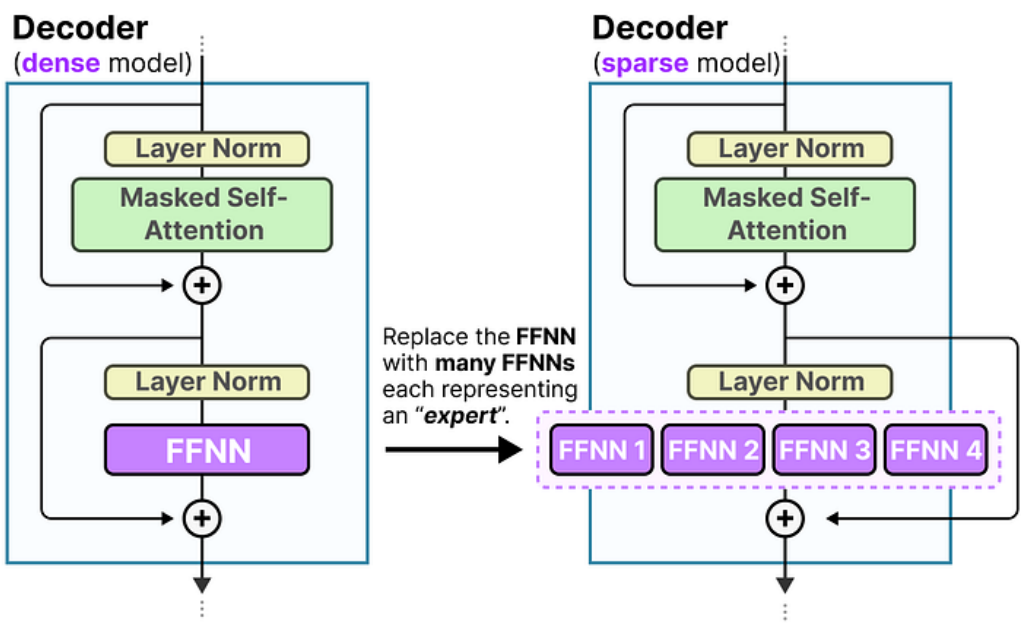
Mixture of Experts enable models to be pretrained with far less compute, which means you can dramatically scale up the model or dataset size with the same compute budget as a dense model.



Source: https://medium.com/@sushma.mareddy/mixture-of-experts-moe-a-simple-explanation-8f7f07ea1439

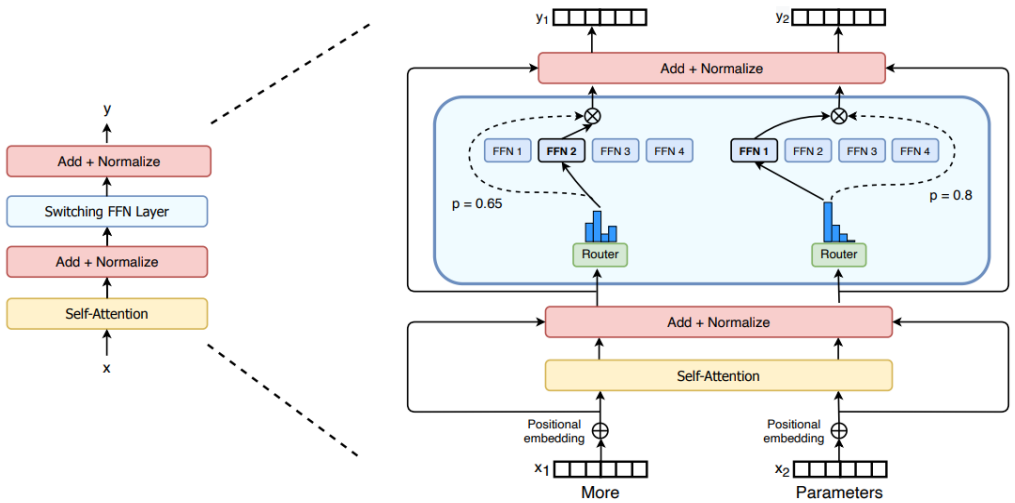
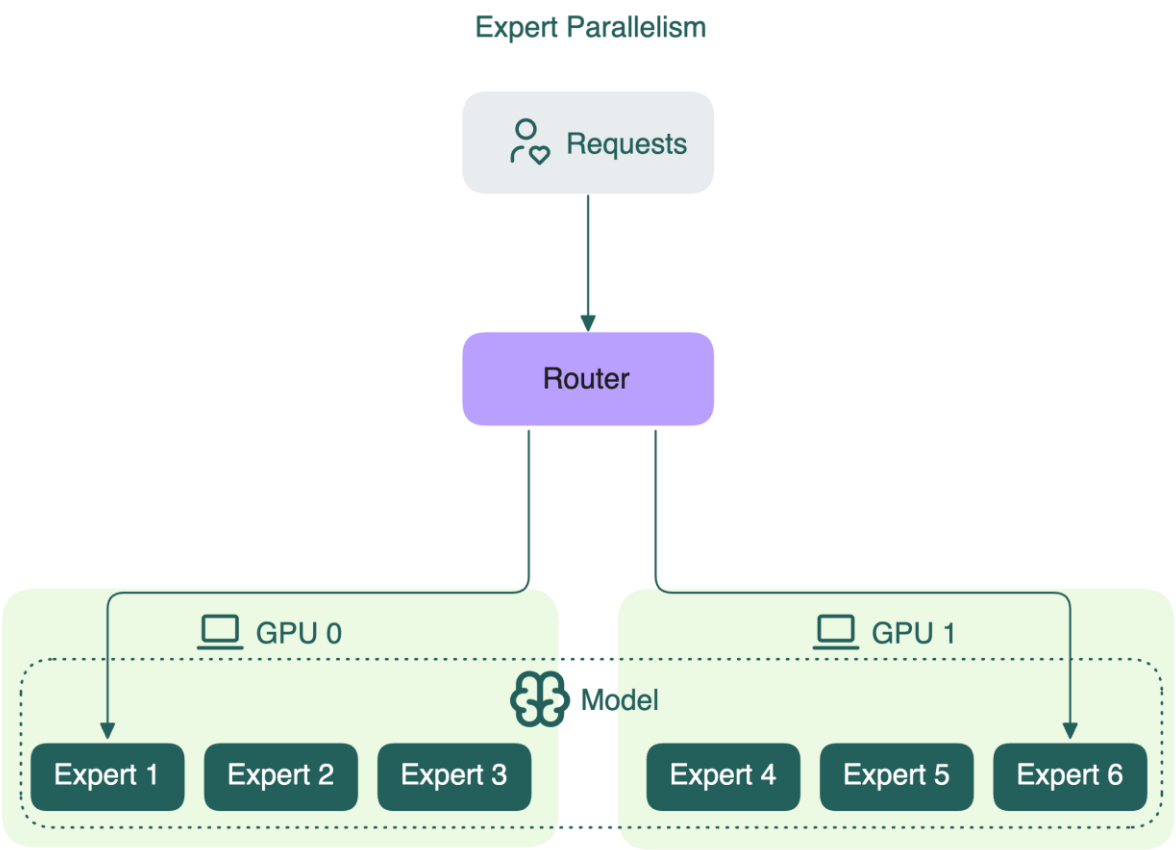


Figure 2: Illustration of a Switch Transformer encoder block. We replace the dense feed forward network (FFN) layer present in the Transformer with a sparse Switch FFN layer (light blue). The layer operates independently on the tokens in the sequence. We diagram two tokens (x_1 = "More" and x_2 = "Parameters" below) being routed (solid lines) across four FFN experts, where the router independently routes each token. The switch FFN layer returns the output of the selected FFN multiplied by the router gate value (dotted-line).

Source: Fedus, William & Zoph, Barret & Shazeer, Noam. (2021). Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. 10.48550/arXiv.2101.03961.

Expert parallelism incurs substantial All-to-All communication overhead



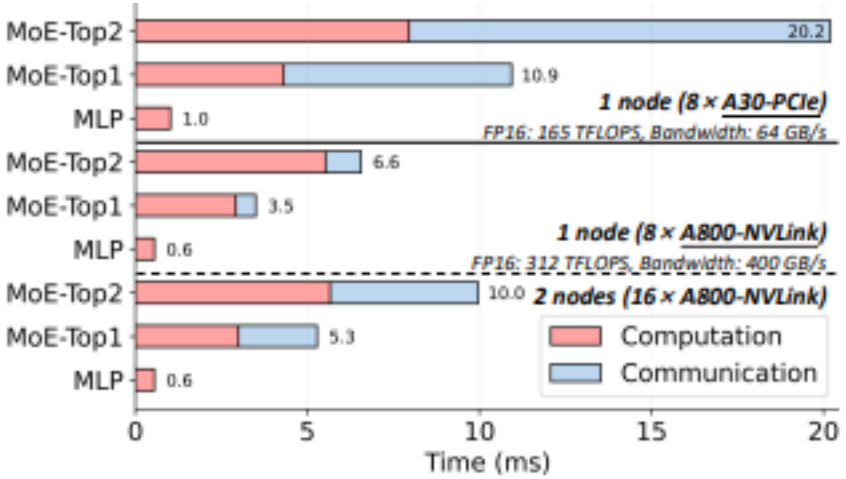


Figure 1. The overhead of MLP and top-2/top-1 MoE in a transformer block of SwinV2-MoE-S (Hwang et al., 2023) model, allocating one expert per GPU with expert parallelism. The All-to-All communication takes up 60% of total time on a single node with 8×A30 GPUs, but drops to 15% on 8×A800 due to the latter's 6× higher bandwidth provided by GPU-to-GPU NVLink (Foley & Danskin, 2017). Despite benefiting from NVLink, communication still approaches 50% due to the lower-bandwidth inter-node Ethernet (Li et al., 2020) when scaling across multiple nodes.

Source: Cai, W., Jiang, J., Qin, L., Cui, J., Kim, S., & Huang, J. (2025). Shortcut-connected Expert Parallelism for Accelerating Mixture of Experts. *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, PMLR 267. arXiv:2404.05019 [cs.LG].

Source: https://bentoml.com/llm/inference-optimization/data-tensor-pipeline-expert-hybrid-parallelism#expert-parallelism

Capacity factor

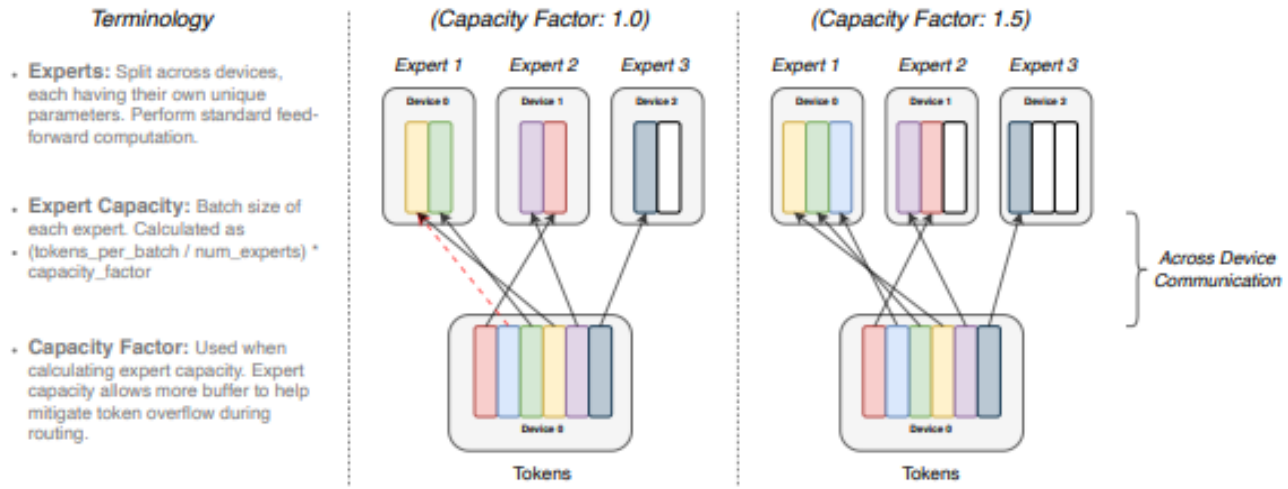


Figure 3: Illustration of token routing dynamics. Each expert processes a fixed batch-size of tokens modulated by the *capacity factor*. Each token is routed to the expert with the highest router probability, but each expert has a fixed batch size of (total_tokens / num_experts) × capacity_factor. If the tokens are unevenly dispatched then certain experts will overflow (denoted by dotted red lines), resulting in these tokens not being processed by this layer. A larger capacity factor alleviates this overflow issue, but also increases computation and communication costs (depicted by padded white/empty slots).

Higher capacity factor means fewer dropped tokens but more wasted compute and memory. Lower means tighter, faster, but you're betting on your router staying balanced.

Source: Fedus, Zoph & Shazeer (2022) *Switch Transformers*, §2.2 "Distributed Switch Implementation." arXiv:2101.03961, JMLR 23(120). CC-BY 4.0.

Git Repo walkthrough and Phase 1 results reporting

https://github.com/Amirjjf/Distributed-MoE-Systems/tree/main/moe-phase1

1/2/4 GPU Comparison table

Run	GPUs	Tokens/sec	GPU Memory (GB)	Expert Imbalance (CV)	Avg Step Time (s)	Scaling Efficiency
1 GPU	1	86,767	2.31	0.036	0.047	100%
2 GPU	2	156,092	2.31	0.030	0.053	89.9%
4 GPU	4	287,968	2.31	0.026	0.057	83.0%

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Kiitos
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